

Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	FAST
Model	qwen3:8b

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Editorial Decision

MAJOR REVISION

Weighted Score: 3.5 / 5.0
Confidence: 84%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	5.0	Excellent	■■■■■
Manuscript Structure	2.5	Weak	■■■■■
Methodology & Equations	4.0	Good	■■■■■
Statistical Analysis	3.5	Good	■■■■■
Results & Evidence	3.0	Acceptable	■■■■■
Discussion & Conclusions	3.5	Good	■■■■■
Figures & Tables	4.0	Good	■■■■■
Scientific Writing	3.5	Good	■■■■■
References & Citations	2.5	Weak	■■■■■
Ethics & Transparency	2.5	Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 5.0 / 5.0 | Confidence: 100%

Strengths:

- [TITLE] The title is specific, informative, and includes key terms for discoverability.
- [ABSTRACT] The abstract follows a structured format with clear background, objective, methods, results, and conclusions.
- [KEYWORDS] The keywords are comprehensive, include domain-specific terms, and avoid duplication of title words.
- [TITLE] The title clearly signals the core method and application domain.
- [ABSTRACT] The abstract provides substantial methodological and validation detail.

StructureReviewer

Manuscript structure and logical coherence

Score: 2.5 / 5.0 | Confidence: 80%

Strengths:

- Clear section mapping
- Logical flow between sections where content is present

Weaknesses:

- Missing literature review section
- Inconsistent content across sections (e.g., repeated content in section_vi and discussion)
- Lack of clear hypothesis or research question
- Section balance is uneven with some sections being disproportionately long or absent

Major Comments:

- The manuscript is missing a critical literature review section, which is essential for establishing the context and significance of the research.
- The content in section_vi and discussion appears to be duplicated, which undermines the logical flow and coherence of the manuscript.
- The hypothesis or research question is not explicitly declared, which is a fundamental requirement for a scientific manuscript.

Minor Comments:

- The section balance is uneven, with some sections being disproportionately long or absent.
- The title and abstract are present but lack clarity and completeness.

Recommendations:

- Include a comprehensive literature review section to contextualize the research.
- Ensure that all sections are logically coherent and free from duplication.
- Clearly state the hypothesis or research question in the introduction.
- Revise the title and abstract to better reflect the content and scope of the manuscript.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 4.0 / 5.0 | Confidence: 80%

Strengths:

- Clear integration of physical constraints into the neural network framework

- Logical flow from problem definition to experimental validation
- Use of a five-story building prototype for validation

Weaknesses:

- Lack of detailed description of the neural network architecture
- No mention of specific optimization algorithms used in the evolutionary search
- No information on the exact experimental setup or data collection process

Major Comments:

- The manuscript should provide more details on the neural network architecture and the specific evolutionary algorithms used (e.g., genetic algorithm, particle swarm optimization).
- The experimental setup and data collection process should be described in more detail to ensure reproducibility.

Minor Comments:

- Equations should be properly introduced and their physical meaning clarified.
- The multi-objective optimization formulation should be more clearly explained.

Recommendations:

- Include a detailed description of the neural network architecture and the evolutionary algorithms used.
- Provide a more comprehensive description of the experimental setup, including data collection and preprocessing steps.
- Clarify the multi-objective optimization formulation and the role of physical constraints in the neural mapping.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Use of Kruskal-Wallis and post-hoc Wilcoxon/Holm corrections for non-parametric comparisons
- Bootstrap confidence intervals for robustness
- RMSE, MSE, and coefficient of variation for performance evaluation
- Pareto/hypervolume indicators for multi-objective optimization

Weaknesses:

- Lack of effect size reporting
- No explicit verification of test assumptions (normality, independence, homoscedasticity)
- No mention of how hypervolume and spread indicators are calculated or interpreted

Major Comments:

- The manuscript should explicitly verify the assumptions of the Kruskal-Wallis and Wilcoxon post-hoc tests (e.g., independence, homoscedasticity) to ensure the validity of the results.
- Effect sizes (e.g., Cohen's d, eta-squared) should be reported alongside p-values to provide a complete picture of the magnitude of differences.
- The interpretation of hypervolume and spread indicators should be clarified to ensure they align with the claims made in the manuscript.

Minor Comments:

- The manuscript should clarify how the coefficient of variation is calculated and interpreted in the context of the study.
- The use of 30 independent runs should be justified in terms of convergence and stability of the results.

Recommendations:

- Include a section on assumption checks for non-parametric tests (e.g., independence, homoscedasticity).
- Report effect sizes (e.g., Cohen's d, eta-squared) to complement p-values.
- Clarify the calculation and interpretation of hypervolume and spread indicators.

- Provide more detail on the justification for using 30 independent runs and how they contribute to the robustness of the results.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: 4.0 / 5.0 | Confidence: 80%

Strengths:

- The manuscript includes a sufficient number of figures, tables, and equations to support the claims made in the text.
- Equations are numbered sequentially and appear to be properly introduced with context.
- Tables and figures are referenced in the text before they appear, which supports clarity and coherence.

Weaknesses:

- The extracted text does not expose table bodies, so it is not possible to verify if the data is precise and consistent across the paper.

Major Comments:

- The manuscript should provide the full content of tables and figures to ensure that the data is precise, consistent, and self-explanatory.

Minor Comments:

- Ensure that all figures and tables are clearly labeled with units and that all variables in equations are defined before use.

Recommendations:

- Include the full content of tables and figures to verify data consistency and completeness.

ResultsReviewer

Results validity and empirical support

Score: 3.0 / 5.0 | Confidence: 80%

Strengths:

- MSE, RMSE, CV, and hypervolume metrics are reported, which are relevant to the multi-objective optimization problem
- Pareto analysis and baseline comparisons with GA and PSO are included
- 30 independent runs are mentioned, which supports robustness analysis

Weaknesses:

- The results section is incomplete and lacks detailed quantitative results, such as specific MSE, RMSE, or hypervolume values
- The comparison with baseline methods is not sufficiently detailed or tied to figures/tables
- Negative results are not reported or acknowledged
- The section ends abruptly without completing the description of the three key aspects (dynamic fidelity, parameter identification, and robustness)

Major Comments:

- The results section is incomplete and lacks detailed quantitative results, such as specific MSE, RMSE, or hypervolume values, which are critical for assessing the performance of the proposed method.

Minor Comments:

- The section ends abruptly without completing the description of the three key aspects (dynamic fidelity, parameter identification, and robustness).
- The baseline comparisons with GA and PSO are not sufficiently detailed or tied to figures/tables.

Recommendations:

- Include specific values for MSE, RMSE, CV, and hypervolume to support the claims made in the results.

- Provide detailed descriptions of the baseline comparisons with GA and PSO, including how they are tied to figures/tables.
- Report negative results honestly and acknowledge any limitations in the results.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **3.5 / 5.0** | Confidence: 90%

Strengths:

- The discussion effectively interprets results in the context of existing methods (e.g., GA and PSO), highlighting the improved robustness and consistency of EANN.
- The conclusions are directly derived from the results, clearly stating the contribution of the EANN framework.

Weaknesses:

- The discussion lacks a detailed comparison with prior work, focusing only on citation without elaborating on how EANN differs or improves upon existing methods.
- The limitations of the study are mentioned but not fully elaborated, such as computational cost and the specific context of the five-story building prototype.
- Future work directions are not specified, which limits the guidance for further research.

Major Comments:

- The discussion should include a more detailed comparison with prior work to strengthen the theoretical implications of the study.

Minor Comments:

- The limitations section could be expanded to provide a clearer understanding of the study's constraints.

Recommendations:

- Include a detailed comparison with existing methods in the discussion to highlight the novelty and practical implications of the EANN framework.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical terminology
- Logical structure of sections

Weaknesses:

- Overuse of passive voice
- Vague quantifiers
- Undefined acronyms

Major Comments:

- Passive voice overuse reduces clarity and engagement.
- Acronyms like EANN should be defined on first use.
- Vague quantifiers like 'significant' weaken the precision of claims.

Minor Comments:

- Some sentences are overly long and could be split for readability.
- Technical terms should be consistently used throughout the manuscript.

Recommendations:

- Define all acronyms on first use.
- Revise passive voice constructions to active voice where possible.

- Replace vague quantifiers with specific numerical values or ranges.
- Break long sentences into shorter, more digestible ones.

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Recent references from 2021-2023
- Consistent citation style (IEEE format)

Weaknesses:

- Lack of seminal foundational works
- Missing key areas in literature review

Major Comments:

- The references section lacks seminal foundational works that are critical to the field of structural health monitoring and lumped mass-spring-damper models. This omission weakens the manuscript's literature review and context.
- The manuscript does not include classic works that are foundational to the field, which is essential for establishing the context and significance of the current research.

Minor Comments:

- The references section includes a mix of conference and journal papers, which is acceptable but could be further balanced for completeness.

Recommendations:

- Include seminal foundational works on structural health monitoring and lumped mass-spring-damper models.
- Ensure the references section includes classic works that are foundational to the field to provide a comprehensive context.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **2.5 / 5.0** | Confidence: 90%

Strengths:

- Limitations are explicitly stated in the discussion section, though they lack specificity.
- The study presents a novel framework for structural parameter identification, which is a significant contribution.

Weaknesses:

- Limitations are not specific or detailed, which reduces the credibility of the findings.
- The manuscript does not address reproducibility by providing data or code availability.
- The absence of ethical approval, conflict of interest, funding, and data availability statements undermines the transparency and rigor of the study.

Major Comments:

- The manuscript lacks essential ethical and transparency information, which is critical for publication in Q1 journals.
- The limitations section is insufficiently detailed and does not adequately address the impact of limitations on the conclusions.

Minor Comments:

- The methodology and results are well-presented, but the lack of reproducibility information is a minor concern.

Recommendations:

- Include a statement on ethical approval for any human or animal studies.
- Provide a data availability statement to ensure transparency and reproducibility.

- Declare any potential conflicts of interest.
- Disclose all funding sources.
- Enhance the limitations section with specific details and their impact on the conclusions.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] The manuscript is missing a critical literature review section, which is essential for establishing the context and significance of the research.
- [2] The content in section_vi and discussion appears to be duplicated, which undermines the logical flow and coherence of the manuscript.
- [3] The hypothesis or research question is not explicitly declared, which is a fundamental requirement for a scientific manuscript.
- [4] The manuscript should provide more details on the neural network architecture and the specific evolutionary algorithms used (e.g., genetic algorithm, particle swarm optimization).
- [5] The experimental setup and data collection process should be described in more detail to ensure reproducibility.
- [6] The manuscript should explicitly verify the assumptions of the Kruskal-Wallis and Wilcoxon post-hoc tests (e.g., independence, homoscedasticity) to ensure the validity of the results.
- [7] Effect sizes (e.g., Cohen's d, eta-squared) should be reported alongside p-values to provide a complete picture of the magnitude of differences.
- [8] The interpretation of hypervolume and spread indicators should be clarified to ensure they align with the claims made in the manuscript.
- [9] The manuscript should provide the full content of tables and figures to ensure that the data is precise, consistent, and self-explanatory.
- [10] The results section is incomplete and lacks detailed quantitative results, such as specific MSE, RMSE, or hypervolume values, which are critical for assessing the performance of the proposed method.
- [11] The discussion should include a more detailed comparison with prior work to strengthen the theoretical implications of the study.
- [12] Passive voice overuse reduces clarity and engagement.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] The section balance is uneven, with some sections being disproportionately long or absent.
- [2] The title and abstract are present but lack clarity and completeness.
- [3] Equations should be properly introduced and their physical meaning clarified.
- [4] The multi-objective optimization formulation should be more clearly explained.
- [5] The manuscript should clarify how the coefficient of variation is calculated and interpreted in the context of the study.
- [6] The use of 30 independent runs should be justified in terms of convergence and stability of the results.
- [7] Ensure that all figures and tables are clearly labeled with units and that all variables in equations are defined before use.
- [8] The section ends abruptly without completing the description of the three key aspects (dynamic fidelity, parameter identification, and robustness).
- [9] The baseline comparisons with GA and PSO are not sufficiently detailed or tied to figures/tables.
- [10] The limitations section could be expanded to provide a clearer understanding of the study's constraints.
- [11] Some sentences are overly long and could be split for readability.
- [12] Technical terms should be consistently used throughout the manuscript.

Specific Improvement Recommendations

1. Include a comprehensive literature review section to contextualize the research.
2. Ensure that all sections are logically coherent and free from duplication.
3. Clearly state the hypothesis or research question in the introduction.
4. Revise the title and abstract to better reflect the content and scope of the manuscript.
5. Include a detailed description of the neural network architecture and the evolutionary algorithms used.
6. Provide a more comprehensive description of the experimental setup, including data collection and preprocessing steps.
7. Clarify the multi-objective optimization formulation and the role of physical constraints in the neural mapping.
8. Include a section on assumption checks for non-parametric tests (e.g., independence, homoscedasticity).
9. Report effect sizes (e.g., Cohen's d, eta-squared) to complement p-values.
10. Clarify the calculation and interpretation of hypervolume and spread indicators.
11. Provide more detail on the justification for using 30 independent runs and how they contribute to the robustness of the results.
12. Include the full content of tables and figures to verify data consistency and completeness.

This review was generated automatically by the Multi-Agent Paper Review System using Phi-3 Mini (Microsoft) running locally via Ollama. No manuscript content was transmitted to external servers. Rubrics based on Elsevier, IEEE, and Emerald Q1 peer review standards. This document does not constitute an official editorial decision.